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Document Control

Version	Date	Author	Description
0.1	20/09/2026	Maria Theresia	Initial draft based on the Customer Support Tickets dataset.
1.0	26/09/2026	Maria Theresia	First complete version including scope, requirements, data dictionary and business questions finalized.

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1. Project Overview

Item	Detail
Project Name	Customer Support Ticket Analytics & Performance Dashboard
Prepared By	Maria Theresia
Role	Business Intelligence Analyst
Tools Used	SQL, Python (Pandas), Power BI, Google Spreadsheet
Dataset	support-tickets.csv containing 4,000 customer support tickets from Jan 2023 – Feb 2024
Document Purpose	Define the business background, problem, objectives, scope, and requirements for analyzing the customer support ticket dataset and building a reporting dashboard

2. Business Background

The Company operates a subscription-based software product and receives customer support requests through four channels: chat, email, in-app messaging, and phone. Support requests are routed into five queues based on topic: Account, Billing, Integrations, Onboarding, and Technical. Common issues include login and password-reset errors, two-factor authentication, billing and invoice questions, data export requests, adding teammates, and third-party data integrations.

Each ticket is logged with a unique ID, the channel and queue it was submitted to, a priority level (low, normal, high, or urgent), the time it was opened, the time to first human reply in minutes (blank if the ticket was never answered), the time it was resolved (blank if the ticket is still open), whether it was reopened after resolution, and a post-resolution customer satisfaction (CSAT) score from 1 to 5, recorded only when a survey was completed.

As the volume of tickets has grown, the Support team has continued to track cases operationally (channel by channel, queue by queue) but has no consolidated, cross-channel view of how the support function is performing as a whole.

3. Business Problem



“The company lacks a centralized view of customer support operations, making it difficult to monitor workload, identify process bottlenecks, and evaluate service performance.”

This is reflected in the current dataset:

- 4,000 tickets were logged between January 2023 and February 2024 across 4 channels and 5 queues, with no single report showing volume, speed, and quality together.
- 225 tickets have no resolution timestamp, so they remain open with no visibility into how this backlog is distributed across queues or priorities.
- First response time ranges from 1 minute to 4,444 minutes (~74 hours), with a median of 83 minutes, indicating high variability that is not currently monitored against any service-level target.
- A CSAT score was captured for only 1,226 of 4,000 tickets (about 31%), and there is no analysis linking satisfaction to response speed, queue, or priority.
- Reopened tickets are recorded per case but not aggregated, so recurring or poorly resolved issues are not systematically identified.

Without a consolidated reporting layer, management cannot answer basic operational questions on demand and instead relies on ad hoc, manual look-ups.

4. Business Objectives

Specific Objective	Expected Business Value
Understand ticket demand: quantify ticket volume, channels, queues, and demand trends over time.	Provides a single, reusable view of support demand, helping managers understand workload distribution and recurring demand patterns.
Identify operational bottlenecks: identify queues, channels, and priority levels where tickets pile up, take longer to resolve, or remain unresolved.	Enables earlier visibility into backlog and slow-moving queues, allowing managers to reallocate workload before service quality is affected.



Monitor response and resolution performance: track first response time and resolution time by priority and queue.	Provides an evidence base for staffing, process, and SLA decisions instead of relying on manual checks or anecdotal observations.
Analyze customer satisfaction: measure CSAT where available and examine its relationship with response speed and other operational factors.	Helps the business understand factors associated with customer satisfaction and identify areas where support performance may need improvement.
Improve support processes through data-driven recommendations: translate findings into actionable recommendations for Support and Operations teams.	Creates a repeatable reporting and improvement framework that can be refreshed as new ticket data becomes available.

5. Project Scope

5.1 In Scope

- Cleaning and structuring the support-tickets.csv dataset (4,000 records, 10 fields).
- Exploratory data analysis of ticket volume, channel mix, queue mix, and priority mix.
- Calculation of first response time and resolution time metrics, and comparison across queue/priority/channel.
- Analysis of unresolved (open) tickets and reopened tickets.
- Analysis of CSAT scores and their relationship with response/resolution speed.
- Design and build of an interactive dashboard/report answering the business questions in Section 8.
- A written summary of insights and recommendations.

5.2 Out of Scope

- Agent-level performance analysis (the dataset does not include an agent/assignee field).
- Predictive modeling or automated ticket routing.
- Real-time/live monitoring: the dashboard is built from a static historical extract.



- Root-cause interviews with customers or support staff.
- Financial/cost analysis of the support operation.
- Any change to production support systems or workflows — this project delivers analysis and recommendations only.

5.3 Milestones

Number	Milestone	Target
1	Data understanding, cleaning & validation	Week 1
2	Exploratory data analysis (volume, speed, quality)	Week 2
3	Dashboard / report build	Week 3
4	Insight synthesis & recommendations	Week 4
5	Final review and portfolio publication	Week 4

5.4 Deliverables

Deliverable	Description
Cleaned dataset	Validated ticket data with derived fields (response time, resolution time, SLA flag, month)
EDA notebook / SQL scripts	Documented exploration of volume, performance, and CSAT
Interactive dashboard	Power BI / Tableau dashboard covering the business questions in Section 8
Business Requirements Document	This document
Insight & recommendation report	Narrative summary of findings and suggested actions

5.5 Acceptance Criteria



- The dashboard/report answers every business question listed in Section 8.
- Reported figures reconcile with the source dataset (row counts, date range, and totals match).
- Missing/incomplete data (unresolved tickets, missing CSAT) is clearly labeled rather than silently excluded.
- The BRD, dashboard, and insight report are internally consistent and reviewed before publication.

6. Stakeholders

Stakeholder	Need
Support Manager	Monitor day-to-day support performance and team workload; primary consumer of the dashboard
Operations Manager	Monitor overall workload and efficiency across queues and channels
Customer Experience Lead	Monitor CSAT trends and customer-impact issues
Business Analyst (Project Author)	Gather requirements, analyze data, and translate business needs into reporting requirements
BI / Data Team	Prepare, validate, and maintain the underlying data and refresh the reporting layer

7. Data Overview

The analysis is based on a single flat file, support-tickets.csv, representing the support ticket log for the period under review.

Attribute	Detail
Records	4,000 tickets
Fields	10 columns
Time period	1 January 2023 – 4 February 2024 (~13 months)



Granularity	One row per ticket
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7.1 Data Dictionary

Field	Description	Type
ticket_id	Unique ticket ID	String
opened_at	Ticket creation time	Datetime
channel	Customer contact channel	String
queue	Support queue	String
priority	Ticket priority	String
subject	Ticket issue	String
first_response_min	First response time	Integer
resolved_at	Resolution timestamp	Datetime
reopened	Reopened status	Boolean
csat	Customer satisfaction	Integer

8. Business Questions

The dashboard and analysis are designed to directly answer the following questions:

1. Which issues (subjects) generate the most tickets?
2. Which queues have the highest workload (ticket volume)?
3. Which channels generate the most tickets?



4. Which queues have slower first-response times?
5. Which priorities require longer resolution times?
6. How does first-response time relate to CSAT?
7. How many tickets are reopened, and where do they concentrate?
8. How many tickets remain unresolved (open), and in which queues/priorities?
9. How does ticket volume trend month over month across the period?

9. Functional Requirements

ID	Priority	Requirement
FR-01	High	Display total ticket volume and trend by month for the full period covered by the dataset.
FR-02	High	Break down ticket volume by channel (chat, email, in-app, phone) and by queue (account, billing, integrations, onboarding, technical).
FR-03	High	Rank the most frequent ticket subjects/issues by volume.
FR-04	High	Calculate average first-response time, segmented by queue and by priority.
FR-05	High	Calculate average resolution time (resolved_at – opened_at), segmented by queue and by priority.
FR-06	Medium	Flag and count tickets with no resolved_at value as "open/unresolved", segmented by queue and priority.
FR-07	Medium	Calculate the reopened ticket rate overall and by queue.
FR-08	Medium	Show CSAT distribution (1–5) and average CSAT by queue and priority, clearly noting the ~31% response coverage.
FR-09	Medium	Show the relationship between first-response time and CSAT score (e.g., average CSAT for fast vs. slow responses).
FR-10	Low	Allow filtering of all views by date range, channel, queue, and priority.



10. Non-Functional Requirements

- The dashboard must load and render the full 4,000-row dataset without noticeable delay on standard business hardware.
- All metrics must be traceable back to the source fields in support-tickets.csv (no black-box calculations).
- Records with missing values (e.g., blank resolved_at or csat) must be visibly labeled, not silently dropped from totals.
- The dashboard must be exportable (PDF/image) for sharing with non-technical stakeholders.
- Visual design should follow a consistent color and labeling convention so priority levels and queues are easy to distinguish at a glance.

11. Assumptions & Constraints

11.1 Assumptions

- The dataset provided (support-tickets.csv) is a complete and accurate export for the stated period; no additional ticket data exists outside this file.
- The absence of an agent/assignee field means agent-level analysis is not possible and is explicitly out of scope.
- CSAT is assumed to be an optional, post-resolution survey, which explains why only ~31% of tickets have a score. This is treated as "missing at random" unless analysis shows otherwise.
- Timestamps are assumed to be in a single, consistent timezone across all channels.
- This is an independent portfolio project; company names, targets, and stakeholder roles are illustrative rather than based on a live business engagement.

11.2 Constraints

- Single analyst working independently, without direct access to customers for follow-up questions.
- Fixed, historical dataset, so no ability to request additional fields (e.g., agent ID, ticket tags, SLA policy) not already present in the file.
- Tooling limited to SQL, Python, Excel, and Power BI.
- Approximately 4 weeks of part-time effort available for the full project.



12. Success Criteria

- All business questions in Section 8 can be answered directly from the delivered dashboard.
- Key figures in the dashboard match manual checks against the raw CSV.
- At least three concrete, data-backed recommendations are produced for improving support operations.
- The finished project (BRD, dashboard, and report) is suitable to present in a portfolio or interview setting.

13. Appendix

13.1 Sample Records

A small illustrative sample from support-tickets.csv (values shown as in the source file):

ticket_id	channel	queue	priority	subject	first_response_min
TK62871	in-app	onboarding	high	Error when log in after password reset	63
TK60343	email	technical	normal	Question about two-factor authentication	7
TK62950	in-app	billing	urgent	Unable to connecting the Snowflake sync	44
TK61014	chat	integrations	normal	Cannot connecting the Snowflake sync	98

13.2 Field Value Reference

Field	Distinct Values
channel	chat, email, in-app, phone
queue	account, billing, integrations, onboarding, technical
priority	low, normal, high, urgent



Business Requirement Documentation

Customer Support Ticket Analytics & Performance Dashboard

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reopened	true, false
csat	1, 2, 3, 4, 5 (populated for ~31% of tickets)